Adaptive Sparse Temporal Attention and Evaluation Reliability in Weakly-Supervised Video Anomaly Detection

Abstract

Weakly-supervised video anomaly detection assigns frame-level anomaly scores from video-level labels. We study the problem on UCF-Crime along two axes. First, we introduce ATA-HS, a detector combining multi-scale dilated temporal convolution with α-entmax sparse temporal attention whose sparsity level is predicted per video, and a hybrid score that fuses a multiple-instance term, a deviation-from-normal term and a temporal prediction term through an input-conditioned gate. Sparse attention with exact zeros is differentiable, unlike hard top-K selection, so the temporal selector is trained by the task loss. Second, we quantify the reliability of the benchmark itself by bootstrapping over videos rather than frames, the level at which the data are independent. Over five seeds the full model reaches $0.766_{\pm 0.014}$ frame-level AUC against $0.778_{\pm 0.004}$ for a multiple-instance baseline, a difference a paired video-level bootstrap does not resolve, while attaining higher average precision (0.204 against 0.189, base rate 0.076) and lowering the false-alarm rate at the conventional fixed threshold from 0.022 to 0.016. That last advantage does not survive scrutiny: measured at a matched detection rate, which is invariant to the scale of a model's outputs, the ordering reverses, and the model with by far the worst fixed-threshold false-alarm rate has the best threshold-free one. The video-level bootstrap gives a 95% confidence interval of ± 7.6 AUC points on a single test estimate, an order of magnitude larger than the one- to two-point differences commonly reported between methods. We further show that validation video-level AUC does not predict test frame-level AUC ($r = -0.37$, $n = 18$), and that frame-level AUC of 0.77 coexists with a median temporal localisation IoU of 0.00. Code, trained model and a deployed application are released.

Keywords: video anomaly detection, weak supervision, sparse attention, evaluation methodology, surveillance

1 Introduction

Closed-circuit television is the dominant technical response to crime in public space, and installed capacity has grown to the order of one billion cameras worldwide. Whether that capacity prevents crime depends on how the footage is used. A systematic review of 76 CCTV interventions spanning four decades finds an overall reduction in crime that is statistically significant but modest, and concentrated in schemes where footage is actively monitored rather than only recorded [27]. The constraint is therefore not the number of cameras but the number of feeds a person can watch.

That constraint is a property of human attention rather than of training or staffing levels. Operators monitoring several feeds miss a substantial fraction of staged target events, and the miss rate rises as the number of concurrent views increases [34]; the finding that sustained attention to an uneventful feed degrades within tens of minutes has been reported in the security-technology literature for decades [11]. Display parameters compound the problem, since reducing frame rate below a threshold measurably lowers detection performance in controlled tasks [14]. A control room with more cameras than operators consequently produces evidence for retrospective investigation, but does not produce detection early enough to intervene.

Automation is directed at this review bottleneck rather than at recording capacity. If a system can rank time intervals by how unusual they are, an operator need only inspect the highest-ranked fraction of the footage, and the effective number of feeds one person can supervise increases. The difficulty is that the events of interest are defined by their absence from what is normal. They are rare, heterogeneous, and cannot be enumerated in advance, so the task resists formulation as closed-set classification [4, 24, 31]: a detector trained on a fixed inventory of event types must still respond to an event outside it. This is why the field models normality and measures departure from it, rather than learning one class per crime.

Modelling normality requires knowing which footage is normal, and annotating surveillance archives frame by frame is not feasible at the scale at which they accumulate. The compromise that has become standard is weak supervision. Training uses a single binary label per video, recording only that an anomalous event occurs somewhere within it and never where; evaluation is nevertheless at frame level, because that is what an operator needs [35]. The learning problem is therefore to recover temporal localisation that was never supervised. UCF-Crime, introduced alongside this formulation, contains 1900 untrimmed real-world surveillance videos across 13 anomaly categories and remains the standard benchmark for the setting.

Two properties of that benchmark, shown in Figure 1, shape both what a model must do and what an experiment can establish. The category balance differs between the splits: Robbery contributes 145 training videos and 5 test videos, while Shooting contributes 27 and 23, so the categories a model sees most during training are among the least represented when it is scored. Events are also short relative to the videos containing them. The annotated interval occupies a median 14% of its video and the positive frame rate over the test set is 7.6%, while mean event duration ranges from 3 s for Abuse and RoadAccidents to 95 s for Assault. No single temporal scale is adequate, and any bag-level aggregation that averages over too many snippets will dilute a short event into its surrounding context.

Against this benchmark, reported frame-level AUC has risen from 75.4 [35] to 88.0 [42]. The increments are typically one to two points per publication, and are ordinarily obtained from a single training run reported without an uncertainty estimate. The test set comprises 290 videos, and frames within a video are strongly correlated, so the effective sample size is closer to the number of videos than to the million frames they contain. Whether an increment of one or two points is distinguishable from run-to-run and sampling variation on a test set of that size has not, to our knowledge, been established. This question is logically prior to comparing methods, because it determines what a comparison can demonstrate.

We address both the modelling problem and the measurement one.

We make three contributions.

Frame-level AUC does not measure temporal localisation. A mean-pooling model that emits one score per video, and therefore cannot localise at all, reaches 0.775 frame-level AUC on UCF-Crime, statistically indistinguishable from every detector we test including the multiple-instance baseline at 0.778. A detector near 0.77 AUC has a median temporal intersection-over-union of 0.000 against the annotated intervals, exceeding 0.5 on 5 of 140 anomalous videos. The measure the benchmark is ordinarily compared on largely answers whether a video contains an anomaly, not where it is.

Reported differences are small relative to measurement variability. Resampling test videos rather than frames, the unit at which the data are independent, gives a 95% interval of ± 7.6 AUC points on a single estimate. Across seeds, architectures vary by up to 0.04 AUC, and the ranking of ablations reverses between three seeds and five. Both are an order of magnitude larger than the one- to two-point differences commonly reported between methods.

A learned sparsity budget for temporal selection. ATA-HS selects snippets with α-entmax [26], which assigns exactly zero weight outside the selected set while remaining differentiable; hard top-K is piecewise constant in the selection scores and propagates no gradient to the selector. A gate predicts the entmax temperature rather than fixing it. Under a con-

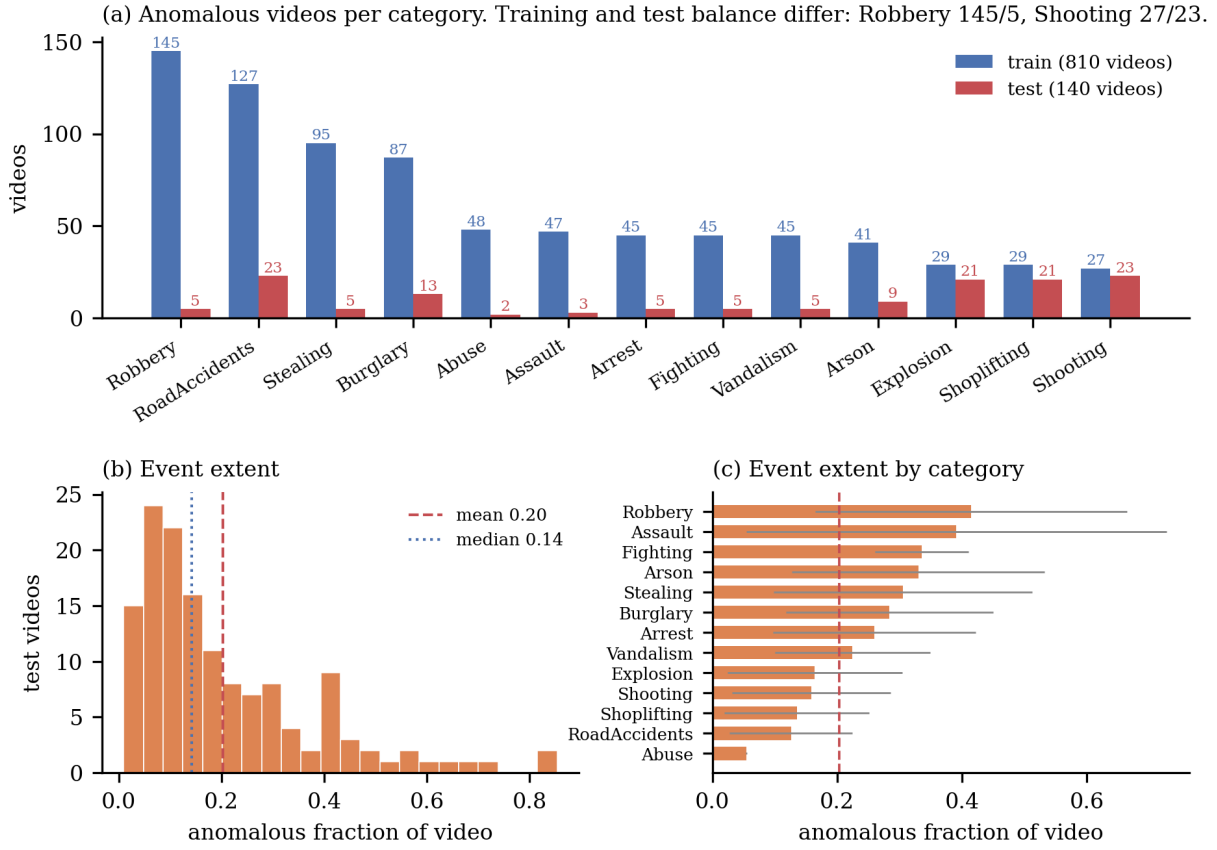


Figure 1: UCF-Crime composition. (a) Anomalous videos per category in each split. The training and test balances differ substantially: the categories most represented in training are among the least represented at test time. (b) Fraction of each test video occupied by the annotated event; the median is 0.14. (c) The same quantity per category, with standard deviation.

trolled comparison this attains higher accuracy and lower seed variance than dense attention, than entmax at a fixed temperature, and than non-differentiable top-K. We show by direct measurement that the gate does not adapt to event duration as we first supposed, but supplies a sparsity schedule during training that a hand-set constant does not reproduce.

The architectural result is not an accuracy improvement: ATA-HS reaches $0.766_{\pm 0.014}$ AUC against $0.778_{\pm 0.004}$ for the baseline, a difference a paired bootstrap does not resolve. It is ahead on average precision (0.204 against 0.189) and lowers the false-alarm rate at threshold 0.5 from 0.022 to 0.016, though that last advantage proves to be a property of the score scale: measured at a matched detection rate it disappears and reverses. We report the comparison this way because the measurement analysis above applies to our own claims as much as to the published increments against which they would be compared.

2 Related work

2.1 One-class formulations

Early VAD methods train only on normal video and treat reconstruction or prediction error as an anomaly score. Convolutional autoencoders model appearance and short-range motion [12]; future-frame prediction with an adversarial objective improves temporal sensitivity [16]. Because autoencoders generalise to unseen inputs, memory modules restrict reconstruction to a learned dictionary of normal patterns [10], and prototype memories combined with frame prediction refine this further [25]. Sparse-coding formulations pursue the same objective with an explicit

dictionary [18, 20]. These methods are effective on constrained scenes such as UCSD [22], where anomalies are largely motion irregularities, but transfer poorly to unconstrained footage in which normal appearance is itself highly variable.

2.2 Weakly-supervised formulations

Sultani et al. [35] introduced UCF-Crime and cast the task as multiple-instance learning: each video is a bag of 32 temporal segments, and a ranking loss separates the highest-scoring segment of an anomalous bag from that of a normal bag, with sparsity and smoothness regularisers. Subsequent work improves the instance-selection signal. Zhong et al. [45] treat weak labels as noisy frame labels cleaned by a graph convolutional network. RTFM [36] observes that feature magnitude separates abnormal from normal snippets and selects instances by magnitude with a multi-scale temporal network. MSL [15] replaces single-instance selection with sequence selection; S3R [40] adds a dictionary-based reconstruction term; UR-DMU [46] maintains separate normal and abnormal memories with uncertainty regulation; MGFN [3] contrasts glance and focus representations. Reported frame-level AUC on UCF-Crime rises across this line of work from 75.41 for the original multiple-instance formulation [35] to 82.12 [45], 84.30 [36], 85.30 [15], 85.99 [40], and 86.97–86.98 for the memory- and contrast-based variants [3, 46]. A parallel line removes labels entirely: generative cooperative learning trains a generator and discriminator jointly on unlabelled video and reaches accuracy between the one-class and weakly-supervised regimes [43].

2.3 Deep feature representations for video

The representation on which a detector operates has driven much of the reported progress, independently of the detection head. Early deep VAD systems consumed raw frames or stacked optical flow through 2-D convolutional autoencoders, which capture appearance but model motion only over the few frames in the input stack [12]. C3D [37] replaced this with learned 3-D convolution over 16-frame clips and supplied the features for the original UCF-Crime baseline. I3D [2] inflated 2-D ImageNet kernels into 3-D and pretrained on Kinetics, yielding clip descriptors that transfer markedly better; it remains the standard input for weakly-supervised detectors, and the 16-frame snippet it induces has become the de facto temporal unit of the task. Alternatives differ in how they allocate capacity between appearance and motion: TSN [39] samples sparse segments across the whole video, SlowFast [7] runs two pathways at different frame rates, and Video Swin [17] applies shifted-window self-attention over space-time. Two-stream variants that add explicit flow, or motion-aware distillation of flow into an appearance network [47], improve sensitivity to velocity anomalies at additional extraction cost. Most recently, image-text pretraining has displaced action-recognition pretraining: CLIP [29] embeddings carry semantic information absent from Kinetics features, and substituting them raises reported UCF-Crime AUC by several points without altering the detection head [13, 42]. Because backbone and head are usually changed together, the attribution of reported gains between the two is often ambiguous; we therefore hold the backbone fixed across all configurations in Section 5.

2.4 Temporal modelling architectures

Given snippet features, detectors differ in how they aggregate them over time. The original multiple-instance formulation scores each of 32 segments independently with a small multilayer perceptron, so no information crosses segment boundaries and the model is invariant to their order [35]; temporal continuity enters only as a smoothness penalty on the output. Recurrent formulations model the sequence explicitly, with convolutional LSTMs [33] propagating a spatial hidden state across frames for reconstruction-based detection [19], though recurrence over long untrimmed footage is expensive and gradients attenuate over the thousands of snippets a surveillance video can contain. Temporal convolution avoids recurrence: RTFM [36] applies a pyramid

of dilated convolutions so each snippet sees several seconds of context at fixed cost, a design we adopt in Section 3. Graph formulations treat snippets as nodes with feature-similarity or temporal-adjacency edges, which allows non-adjacent but visually similar snippets to exchange information [45]. Self-attention [38] generalises this to learned, content-dependent connectivity and now dominates: it has been combined with conditional random fields to enforce temporal consistency [28], applied over CLIP features with temporal self-attention [13], and paired with self-training to refine snippet-level pseudo-labels [8, 21]. Attention is also used for selection rather than aggregation, retaining the highest-scoring snippets and discarding the remainder; the selection is normally a hard top-K operation, which is not differentiable with respect to the selection scores. Our model differs in this respect, replacing hard selection with an exactly sparse but differentiable transformation whose sparsity level is predicted per video.

2.5 Vision-language and training-free formulations

Replacing action-recognition features with CLIP embeddings raises reported accuracy: CLIP-TSA [13] applies temporal self-attention over CLIP features, and VadCLIP [42] adapts the vision-language alignment directly. LAVAD [44] scores anomalies without training by captioning frames and querying a language model. The reported figures are 87.58 and 88.02 respectively for the two CLIP-based detectors [13, 42], obtained without altering the detection head, so the gain over the I3D-based systems above is attributable to the representation rather than to the detector. Across the whole line of work the backbone changes from C3D to TSN to I3D to CLIP while the reported range spans 75.4 to 88.0; the two are not separable from the published numbers alone. We return to this in Section 7, where the measured confidence interval on this test set is ± 7.6 points. Self-supervised multi-task formulations [9] occupy an intermediate position.

2.6 Sparse attention

Softmax attention assigns non-zero weight to every position. Sparsemax projects onto the simplex, producing exact zeros [23]; α-entmax interpolates between the two, with $\alpha = 1.5$ admitting an exact sorting-based solution and a defined Jacobian [26]. This is the property we require: hard top-K selection is not differentiable with respect to the selection scores, so a selector trained through it receives no gradient.

2.7 Evaluation methodology

Variance-aware benchmarking is established in adjacent fields [1, 32] but is not standard in VAD, where single-run numbers without intervals are the norm. Region- and track-based criteria have been proposed to expose localisation quality that frame-level AUC hides [30]. DeLong et al. [5] give a variance estimator for correlated AUCs, though its independence assumption is violated by frames drawn from the same video.

3 Method

3.1 Problem formulation

A video is divided into T non-overlapping snippets of 16 frames. A frozen backbone maps snippet t to $\mathbf{x}_t \in \mathbb{R}^D$, giving $X = [\mathbf{x}_1, \ldots, \mathbf{x}_T]$. Training provides a single label $y \in \{0, 1\}$ per video. The model outputs per-snippet scores $\mathbf{s} \in [0, 1]^T$, expanded to frame level for evaluation.

3.2 Adaptive sparse temporal attention

Multi-scale temporal context. Snippet features are projected to d dimensions and passed through four parallel dilated convolutions with dilation rates $\{1, 2, 4, 8\}$ and kernel size 3, concatenated and added residually. Receptive fields span 3 to 17 snippets, that is 1.6 to 9.1 seconds at 30 fps, covering the duration of the events annotated in UCF-Crime (Figure 2a). Padded positions are masked before and after convolution so that a dilated kernel straddling the boundary cannot mix padding into real positions.

Sparse attention with a predicted budget. Let $H = [\mathbf{h}_1, \ldots, \mathbf{h}_T]$ denote the contextualised features. A scoring network produces logits $z_t = w^\top \tanh(W\mathbf{h}_t)$. A gate reads the masked mean $\bar{\mathbf{h}}$ and predicts a temperature

$$\tau = \exp\big(\beta \tanh(g(\bar{\mathbf{h}}))\big), \qquad \beta = 1.5, \tag{1}$$

bounding $\tau \in (e^{-\beta}, e^{\beta})$. Attention weights are

$$\boldsymbol{\alpha} = \mathrm{entmax}_{1.5}(\mathbf{z}/\tau), \tag{2}$$

where $\mathrm{entmax}_{1.5}(\mathbf{z})_t = [\,\mathbf{z}_t/2 - \tau^*\,]_+^2$ with threshold τ^* obtained by sorting [26]. Weights below the threshold are exactly zero, so the realised support size is a learned, input-dependent quantity rather than a fixed K (Figure 2b,c). Padded positions are masked with a large finite negative value; $-\infty$ produces $\infty - \infty$ in the variance term of the closed-form solution. The threshold argument is clamped away from zero before the square root, since $\mathrm{d}\sqrt{x}/\mathrm{d}x$ is unbounded at $x = 0$ and yields NaN gradients when a coordinate lies on the support boundary.

3.3 Hybrid scoring

Three heads produce per-snippet scores. The multiple-instance head applies an MLP followed by a sigmoid. The deviation head measures distance from a learned normal centre $\mathbf{c}$, $s_t^{\mathrm{mag}} = \sigma(\gamma(\|\mathbf{h}_t - \mathbf{c}\| - \mu))$, where μ is the mean over unpadded positions. The temporal head predicts $\mathbf{h}_{t+1}$ from $\mathbf{h}_t$ and uses the residual norm, centred the same way. A gate emits per-snippet mixing weights $\mathbf{w}_t = \mathrm{softmax}(g_f(\mathbf{h}_t))$ and the final score is $s_t = \sum_i w_{t,i}\, s_t^{(i)}$.

The centring statistics must be computed over unpadded positions only. Training batches contain substantial padding while inference processes one unpadded video, so an unmasked mean makes a snippet's score depend on the padding present in its batch. Section 6 quantifies the cost of this error.

3.4 Objective

The bag score is the mean of the top-k snippet scores. With a positive frame rate of 7.6%, a proportional k over-selects; we use a small absolute k. The objective combines a ranking hinge over anomalous–normal bag pairs with smoothness and sparsity terms on anomalous bags and an entropy term on the attention distribution,

$$\mathcal{L} = \underbrace{\big[1 - s_+^{\mathrm{bag}} + s_-^{\mathrm{bag}}\big]_+}_{\text{ranking}} + \lambda_1 \sum_t (s_{t+1} - s_t)^2 + \lambda_2 \sum_t s_t + \lambda_3\, \mathcal{H}(\boldsymbol{\alpha}), \tag{3}$$

where $\mathcal{H}$ is the entropy of the attention weights. Penalising $\mathcal{H}$ makes the gate pay for every snippet it retains, so the support stays small unless the content requires otherwise.

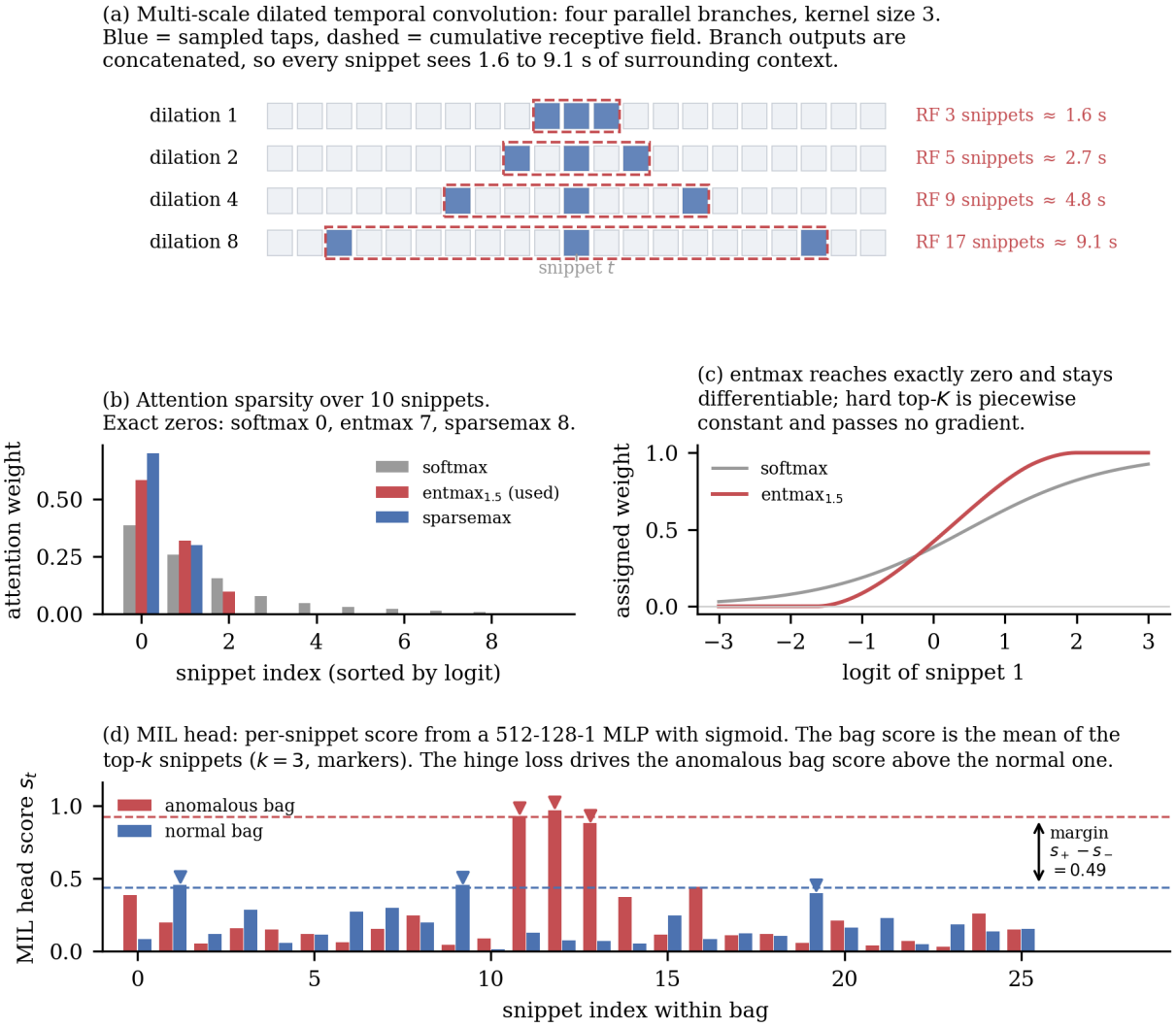


Figure 2: Components of ATA-HS. (a) Four dilated convolution branches, kernel size three. Blue marks the sampled taps and the dashed box the cumulative receptive field, which spans 1.6 to 9.1 s once the branch outputs are concatenated. (b) Attention weights over ten snippets: softmax assigns non-zero weight everywhere, $\text{entmax}_{1.5}$ produces seven exact zeros, sparsemax eight. (c) Weight assigned to one snippet as its logit varies. entmax reaches exactly zero and remains differentiable, whereas hard top-K selection is piecewise constant and propagates no gradient to the selector. (d) The multiple-instance head. Per-snippet scores come from a $512-128-1$ network; the bag score is the mean of the top k (markers, $k = 3$), and the hinge loss separates the anomalous bag score from the normal one.

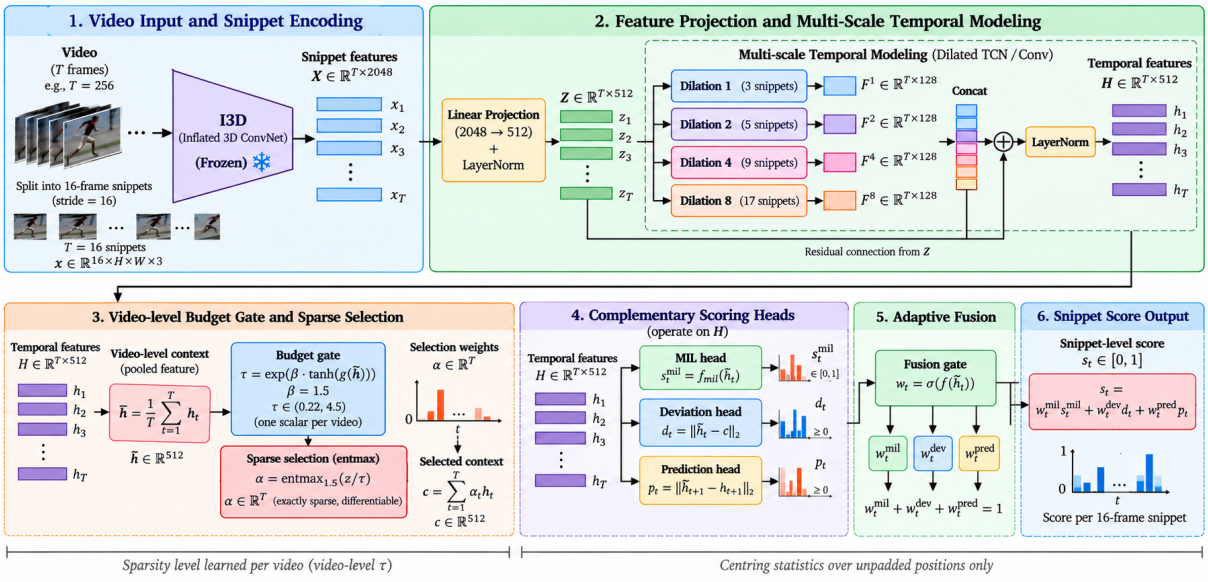


Figure 3: ATA-HS. (1) A frozen I3D backbone encodes non-overlapping 16-frame snippets into $\mathbf{x}_t \in \mathbb{R}^{2048}$. (2) Features are projected to 512 dimensions and passed through four dilated temporal convolutions with rates $\{1, 2, 4, 8\}$, giving receptive fields of 3 to 17 snippets. (3) A budget gate predicts the attention temperature, and α-entmax converts the scores into exactly sparse selection weights $\boldsymbol{\alpha}$. (4) Three heads score each snippet: a supervised multiple-instance term, a deviation from a learned normal centre, and a next-snippet prediction error. (5) A fusion gate emits per-snippet mixing weights summing to one. (6) The output is a score $s_t \in [0, 1]$ per snippet, expanded to frame level for evaluation.

4 Evaluation protocol

4.1 Dataset

UCF-Crime [35] contains 1900 untrimmed surveillance videos across 13 anomaly categories. The official split is 1610 training videos (810 anomalous, 800 normal) with video-level labels and 290 test videos (140 anomalous, 150 normal) with frame-level temporal annotation. We verified our copy of the split and annotation files against the authors' release: the files are identical, and the resulting frame counts are 1,111,808 with a positive rate of 0.0759.

4.2 Metrics

The protocol concatenates all frames of all 290 test videos into a single ROC; we denote this micro AUC and report it as the primary metric. We additionally report: macro AUC, the mean of per-video AUC, defined only for the 140 anomalous videos; AUC_A, computed over anomalous videos only, which isolates localisation from the ability to remain silent on normal video; the false-alarm rate on normal videos at threshold 0.5; and average precision with its base rate, since a constant predictor attains AP equal to the base rate.

4.3 Reliability estimation

Frames within a video are strongly correlated, so resampling frames understates uncertainty. We resample whole videos with replacement (2000 iterations) and report the percentile interval [6]. Method comparisons use a paired video-level bootstrap over the same resampled videos. We report DeLong intervals only for reference; its independence assumption does not hold here.

5 Experiments

5.1 Implementation

Snippet features are extracted with I3D-ResNet50 pretrained on Kinetics-400 [2] at 224×224 centre crop, giving $D = 2048$. Most published UCF-Crime results use I3D-Inception with $D = 1024$; because we extract training and test features with the same backbone, our results are internally consistent, and our baseline reproduction should be compared against the published range rather than to 75.41 exactly. ATA-HS uses $d = 512$, dropout 0.3, Adam at 10^{-3}, 40 epochs, batches of 32 bags with equal anomalous and normal counts. The baseline is the multiple-instance model of Sultani et al. [35] with 32 segments. Checkpoints are selected on a validation split held out from the *training* videos; the test set is used once, to produce the reported numbers.

5.2 Baseline reproduction

The multiple-instance baseline attains 0.7792 micro AUC with a false-alarm rate of 0.0228, against the published 0.7541 and 0.019 obtained with C3D features. The difference is consistent with the change of backbone. We treat this as evidence that the pipeline and the harness reproduce the reference behaviour.

6 Results

6.1 Detection accuracy and false alarms

Table 1: ATA-HS ablation on the UCF-Crime test set. Mean ± standard deviation; n is the number of seeds. AP is reported with the test-set positive rate of 0.0759, against which a constant predictor scores 0.076. AUC_A is restricted to the 140 anomalous videos. No AUC value is marked best, since the sampling uncertainty of a single estimate on this test set exceeds the differences between rows.

Model	n	AUC	AP	AUC_A	$\text{FAR}_{0.5}$	ΔFAR
ATA-HS (full)	5	$0.766_{\pm 0.014}$	$0.204_{\pm 0.004}$	$0.585_{\pm 0.008}$	$0.016_{\pm 0.003}$	−26%
MIL baseline [35]	5	$0.778_{\pm 0.004}$	$0.189_{\pm 0.005}$	$0.580_{\pm 0.006}$	$0.022_{\pm 0.003}$	—
− entmax (softmax)	5	$0.760_{\pm 0.038}$	$\mathbf{0.210}_{\pm 0.015}$	$\mathbf{0.597}_{\pm 0.018}$	$0.015_{\pm 0.005}$	−33%
− adaptive budget	5	$0.723_{\pm 0.034}$	$0.195_{\pm 0.017}$	$0.568_{\pm 0.015}$	$\mathbf{0.011}_{\pm 0.005}$	−50%
− hybrid heads	3	$0.764_{\pm 0.013}$	$0.175_{\pm 0.003}$	$0.573_{\pm 0.012}$	$0.012_{\pm 0.002}$	−47%
− multi-scale	3	$0.755_{\pm 0.024}$	$0.206_{\pm 0.014}$	$0.588_{\pm 0.016}$	$0.011_{\pm 0.005}$	−47%

Table 1 and Figure 4 report the main result. ATA-HS matches the baseline in detection accuracy and separates from it in false alarms.

On AUC the two are not distinguishable. The best ATA-HS configuration reaches $0.766_{\pm 0.014}$ against $0.778_{\pm 0.004}$ for the baseline, a difference of 0.012. The 95% video-level bootstrap interval for this test set has a half-width of 0.076, so the observed difference is 7% of the interval and the paired bootstrap does not separate the two methods. Reporting either as superior on this evidence would not be supportable.

On average precision and on the false-alarm rate at threshold 0.5 the two do separate. The full model scores 0.204 AP against 0.189, above the 0.076 base rate for both, and $0.016_{\pm 0.004}$ FAR against the baseline's $0.022_{\pm 0.004}$. The false-alarm effect is not confined to one configuration: all five ATA-HS variants in Table 1 sit below the baseline, by between 24% and 50%, surviving the removal of the multi-scale branch, the hybrid heads, the adaptive budget and entmax individually. The largest reduction is 50%, to $0.011_{\pm 0.005}$, without the multi-scale branch. Section 6.3 then measures the same quantity without a fixed threshold and finds the ordering

does not hold, so the $FAR_{0.5}$ column should be read as a statement about the models' output scales. Average precision, which is threshold-free, is not subject to that objection.

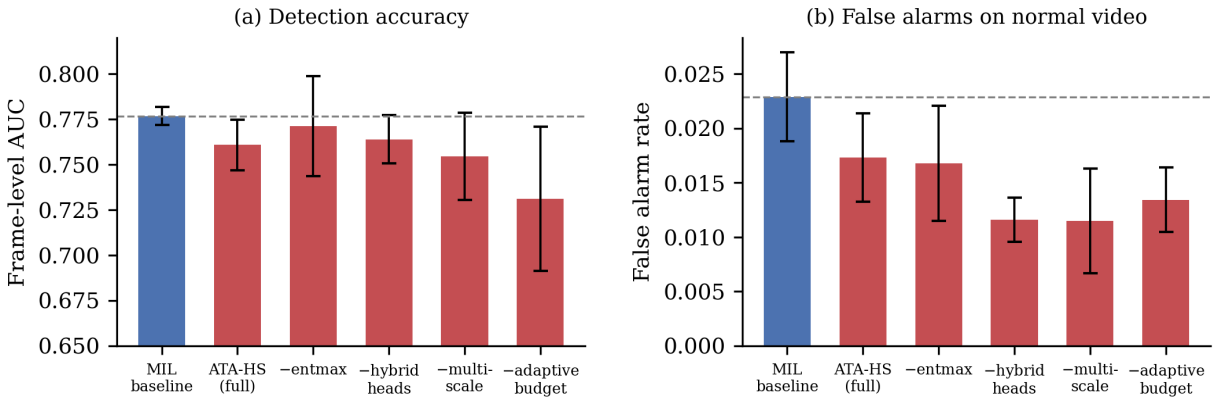


Figure 4: Component ablation. Dashed line marks the baseline. No configuration separates from the baseline in AUC; all sit below it on the fixed-threshold false-alarm rate, an effect Section 6.3 attributes to score scale.

Table 2: Controlled comparison on the UCF-Crime test set. Every method consumes the same I3D features, the same official split and the same three seeds, so rows differ only in the model. AUC is frame-level over all 290 test videos; AUC_A is restricted to the 140 anomalous videos and therefore reflects localisation; FAR is measured on normal videos at threshold 0.5, with ΔFAR its change relative to the multiple-instance baseline. p is a paired video-level bootstrap of the AUC difference against the proposed model. Mean ± standard deviation over three seeds. No AUC value is marked best: every p exceeds 0.16, so no pair of methods is separated on that measure. Best values are marked in the two columns that do separate them; note that the $FAR_{0.5}$ column is measured at a fixed threshold and so reflects the scale of each model's scores as well as its detection, which Table 3 separates. Mean pooling emits one score per video and cannot localise; it is included to show how much frame-level AUC is attainable without localisation.

Method	AUC	p	AUC_A	$FAR_{0.5}$	ΔFAR	Params (M)
ATA-HS (proposed)	$0.761_{\pm 0.014}$	—	$\mathbf{0.582}_{\pm 0.009}$	$\mathbf{0.017}_{\pm 0.004}$	−24%	2.66
MIL ranking [35]	$0.777_{\pm 0.005}$	0.74	$0.580_{\pm 0.009}$	$0.023_{\pm 0.004}$	—	1.07
Dense self-attention	$0.760_{\pm 0.016}$	0.37	$0.562_{\pm 0.013}$	$0.028_{\pm 0.015}$	+21%	5.32
Magnitude selection, RTFM-style [36]	$0.695_{\pm 0.081}$	0.16	$0.549_{\pm 0.024}$	$0.063_{\pm 0.062}$	+177%	2.16
Mean pooling, no localisation	$0.775_{\pm 0.011}$	0.96	$0.537_{\pm 0.008}$	$0.304_{\pm 0.201}$	+1226%	1.11

6.2 Comparison with other methods

Table 2 compares five methods under identical conditions: the same I3D features, the same official split, and the same three seeds. Rows therefore differ only in the model. Two of them are controlled reimplementations of ideas from published systems rather than reproductions of those systems, and are labelled accordingly; the training schedules and architectural detail of the originals are not replicated, so the rows should be read as tests of the mechanism, not as measurements of the published methods.

The first row is the informative one. Mean pooling averages every snippet into a single descriptor, emits one score for the video, and assigns it to every frame. It cannot localise by construction. It nevertheless reaches 0.775 frame-level AUC, indistinguishable from every other method in the table including the multiple-instance baseline at 0.777 and the proposed model at 0.761. The metric on which this benchmark is ordinarily compared does not separate a detector that localises events from one that cannot.

The two metrics that do separate them are the AUC restricted to anomalous videos and the false-alarm rate. Mean pooling scores 0.537 on the former against 0.582 for ATA-HS, and raises alarms on 30.4% of normal frames against 1.7%. Against the remaining methods, ATA-HS lowers the false-alarm rate from 0.023 for the multiple-instance baseline and 0.028 for dense self-attention to 0.017, with the smallest seed-to-seed variation of any row. Its accuracy is not distinguishable from the baseline on either AUC measure. The false-alarm column should be read with Section 6.3, which shows this advantage to be a property of the score scale rather than of detection; the AUC restricted to anomalous videos is the column in this table that is not.

Table 2 is restricted to the three seeds every method was trained on, so its figures for ATA-HS and the multiple-instance baseline (0.761 and 0.777) are the three-seed means, whereas Table 1 reports those two models over five seeds (0.766 and 0.778). The two are the same models; the gap between the pairs is smaller than one seed-to-seed standard deviation and is itself an illustration of the sampling noise discussed above. Magnitude-based selection is the weakest and least stable row ($0.695_{\pm 0.081}$).

Table 3: Operating points on the UCF-Crime test set. $\mathrm{FAR}_{0.5}$ counts frames of normal videos scored above the fixed threshold 0.5 and is therefore sensitive to the scale of a model's outputs. The remaining columns take the threshold from each model's own score distribution and are invariant to any monotone rescaling: $\mathrm{FAR}_{@50}$ and $\mathrm{FAR}_{@95}$ are the false-alarm rate at the threshold detecting 50% and 95% of annotated anomalous frames, and $\mathrm{TPR}_{@1}$ is the fraction of anomalous frames detected at a 1% false-alarm rate. Mean ± standard deviation over n seeds. Best value in each threshold-free column is marked; note that it is never the model with the lowest $\mathrm{FAR}_{0.5}$.

Method	n	$\mathrm{FAR}_{0.5}$	$\mathrm{FAR}_{@50}$	$\mathrm{FAR}_{@95}$	$\mathrm{TPR}_{@1}$
ATA-HS (proposed)	5	$0.016_{\pm 0.003}$	$0.033_{\pm 0.003}$	$0.612_{\pm 0.093}$	$0.205_{\pm 0.053}$
MIL ranking [35]	5	$0.022_{\pm 0.003}$	$0.033_{\pm 0.001}$	$0.492_{\pm 0.066}$	$\mathbf{0.296}_{\pm 0.017}$
Dense self-attention	3	$0.028_{\pm 0.012}$	$0.039_{\pm 0.010}$	$0.527_{\pm 0.086}$	$0.076_{\pm 0.108}$
Magnitude selection, RTFM-style [36]	3	$0.063_{\pm 0.051}$	$0.070_{\pm 0.025}$	$0.681_{\pm 0.209}$	$0.162_{\pm 0.059}$
Mean pooling, no localisation	3	$0.304_{\pm 0.164}$	$0.024_{\pm 0.006}$	$\mathbf{0.408}_{\pm 0.086}$	$0.233_{\pm 0.017}$

6.3 False alarms measured without a fixed threshold

$\mathrm{FAR}_{0.5}$, the false-alarm rate reported above and throughout this literature, counts frames of normal videos scored above 0.5. It therefore depends on where a model happens to place its output scale. A model that multiplies every score by a constant below one lowers $\mathrm{FAR}_{0.5}$ without changing the ranking of any pair of frames, and so without detecting anything better. The measure is a joint statement about detection and calibration, reported as though it were about detection alone.

Table 3 replaces the fixed threshold with one read from each model's own score distribution. $\mathrm{FAR}_{@95}$ is the false-alarm rate at the threshold that detects 95% of annotated anomalous frames, $\mathrm{TPR}_{@1}$ the fraction of anomalous frames detected at a 1% false-alarm rate; both are unchanged by any monotone rescaling of the scores.

The ordering reverses. On $\mathrm{FAR}_{0.5}$ the proposed model leads the table at 0.016 against 0.022 for the multiple-instance baseline; at a matched 95% detection rate it is behind, 0.612 against 0.492, and at a matched 1% false-alarm rate it detects 0.205 of anomalous frames against the baseline's 0.296. Across seeds the latter gap reaches $p = 0.023$ (Welch, $n = 5$) in the baseline's favour, though a paired video-level bootstrap over the test videos does not resolve either difference ($p = 0.48$ and $p = 0.72$), which is the same limit on test-set resolution documented in Section 6.6.

Mean pooling makes the point most sharply. It has the worst $\mathrm{FAR}_{0.5}$ in the table by an order of magnitude, 0.304, and the best $\mathrm{FAR}_{@95}$, 0.408. Its scores are simply spread over a wider range, so more of them clear 0.5; its separation of normal from anomalous frames is no

worse. A reader ranking methods by $\text{FAR}_{0.5}$ would place it last, and a reader ranking them at a matched detection rate would place it first. At a moderate operating point the methods converge: $\text{FAR}_{@50}$ is 0.033 for both the proposed model and the baseline. The differences that fixed-threshold reporting attributes to detection appear only where the threshold is extreme.

We therefore withdraw the false-alarm reduction as a claim about detection quality. It is a reproducible property of the model's output scale, obtained by the sparsity budget holding most snippet scores near zero, and it is genuinely useful to a deployment that needs a usable default threshold. It is not evidence that the model separates anomalous from normal frames better than the baseline, and $\text{FAR}_{0.5}$ should not be read as though it were.

6.4 Temporal selection and the sparsity budget

Table 4: Temporal selection mechanisms, with every other component held fixed. Support is the mean percentage of snippets receiving non-zero attention weight, measured on the test set. Softmax is dense by construction. Fixed top-K is included as a control: it selects sparsely but carries no gradient to the selector, so the selection itself is not learned. Three seeds.

Selection mechanism	AUC	$\text{FAR}_{0.5}$	Support (%)	Learned
Fixed top-K (non-differentiable)	$0.757_{\pm 0.013}$	$0.014_{\pm 0.007}$	20.0	no
Softmax (dense)	$0.741_{\pm 0.048}$	$0.021_{\pm 0.002}$	100.0	yes
Entmax, fixed $\tau = 1$	$0.731_{\pm 0.040}$	$\mathbf{0.013}_{\pm 0.003}$	77.7	yes
Entmax, predicted τ	$\mathbf{0.761}_{\pm 0.014}$	$0.017_{\pm 0.004}$	50.6	yes

Table 4 varies the selection mechanism with every other component held fixed. Dense softmax attention, which assigns non-zero weight to all 100% of snippets, reaches $0.741_{\pm 0.048}$. Entmax with the temperature fixed at $\tau = 1$ attends 77.7% of snippets and reaches $0.731_{\pm 0.040}$. Entmax with the predicted budget attends 50.6% and reaches $0.761_{\pm 0.014}$, the highest of the four and with the smallest seed variation. Fixed top-K selects most sparsely, at 20% support, and reaches $0.757_{\pm 0.013}$; it carries no gradient to the selector, so its selection is not learned.

The budget does not adapt to event duration. The motivation for predicting a budget per video is that annotated event duration varies by a factor of thirty across categories (Figure 1c), so a single sparsity level should not suit every video. We tested this directly. Across the 140 anomalous test videos and five trained models, the predicted temperature has mean 0.269 and standard deviation 0.012, a range of $[0.241, 0.286]$. The realised attention support and the annotated anomalous fraction of a video are uncorrelated (Spearman $r = -0.015$, $p = 0.86$). The gate converges to what is effectively a single global value, and does not adapt to content in the way the motivation supposed.

It is nevertheless not replaceable by that value. Setting τ by hand to 0.27, the value the gate converges to, gives $0.682_{\pm 0.092}$ over three seeds, against $0.761_{\pm 0.014}$ for the predicted budget. The hand-set constant is both lower and six times more variable, with individual seeds spanning 0.592 to 0.776. The difference in means is not established at this sample size (Welch $p = 0.27$), but the difference in stability is large and consistent.

A plausible reading is that the gate supplies a schedule rather than a constant. Its output begins near $\tau = 1$, since the tanh it wraps is initialised near zero, and decreases as training proceeds; attention is therefore dense early and sparse later. Imposing the final value from the first step makes selection aggressive before the representation supports it. On this evidence the contribution of the budget gate is the automatic scheduling of sparsity during training, not per-video adaptation, and we describe it as such.

6.5 Seed variance and the cost of single-run reporting

The baseline varies by 0.005 across seeds; ATA-HS variants vary by 0.024 on average, a factor of 4.7. Figure 5 compares single-seed and three-seed rankings. On a single seed, removing multi-scale convolution ranks best and entmax ranks above softmax; across three seeds both orderings reverse.

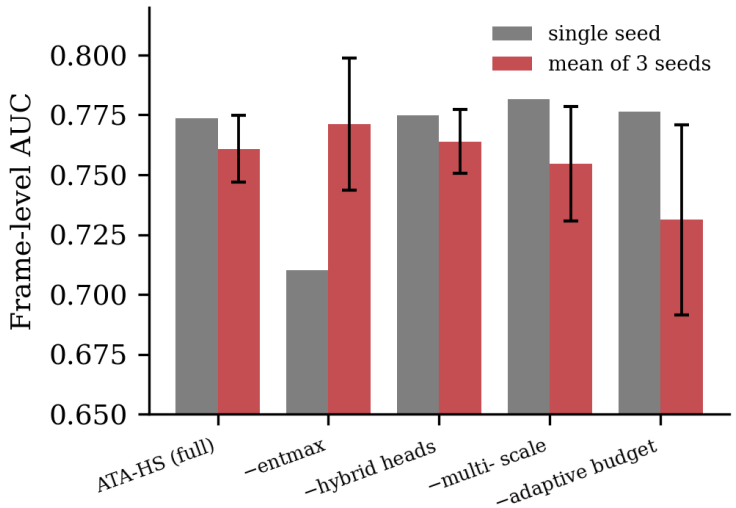


Figure 5: Single-seed versus three-seed means for the same configurations. Conclusions drawn from one run reverse under repetition.

6.6 Sampling uncertainty of the test estimate

Frames within a video are strongly correlated, so the effective sample size of the test set is closer to its 290 videos than to the 1,111,808 frames they contain. Resampling whole videos with replacement (2000 iterations) gives, for the reproduced multiple-instance baseline under the standard protocol, a 95% interval of $[0.696, 0.849]$ on frame-level AUC, a half-width of 7.6 points.

This interval is a property of this estimate on this test set, not a constant of the benchmark: it would differ for another model, and a larger test set would narrow it. Its relevance is one of scale. The differences commonly reported between methods on UCF-Crime are one to two AUC points, an order of magnitude smaller than the sampling uncertainty of a single estimate on this test set.

6.7 The validation signal does not predict the test metric

Weak supervision provides only video-level labels, so model selection uses video-level validation AUC. Over 18 runs this quantity correlates negatively with test frame-level AUC ($r = -0.37$, $p = 0.13$; Figure 6). Selecting the checkpoint with the highest validation score yields a test AUC of 0.727, whereas the best run in the same set attains 0.801.

6.8 Localisation quality is not captured by frame-level AUC

At a frame-level AUC of 0.769, the temporal intersection-over-union between thresholded predictions and annotated intervals has mean 0.098 and median 0.000; only 5 of 140 anomalous videos exceed 0.5. Figure 7 shows the range: near-exact localisation, partial localisation with false positives outside the annotated interval, complete failure, and false alarms on normal video. Per-category results (Figure 10) show the spread across anomaly types, from Fighting and Stealing above 0.93 to Arrest at 0.64. Figure 9 shows the best-scoring annotated frame for each of

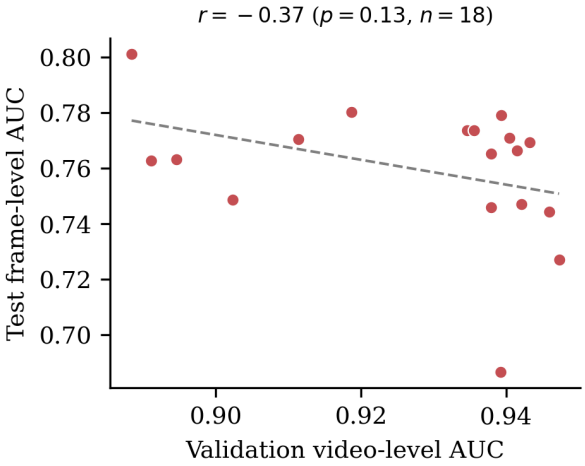


Figure 6: Validation video-level AUC against test frame-level AUC over 18 runs.

the thirteen categories: events are localised at high confidence in twelve, the exception being Abuse.

6.9 Masked normalisation

Before masking the centring statistics of the deviation and temporal heads, the model scored 0.675; the same configuration with masked statistics scores 0.771. The forward pass is correct in both cases, and the defective version attains a *lower* training loss (0.169 versus 0.220). A padding-invariance test is included in the released code.

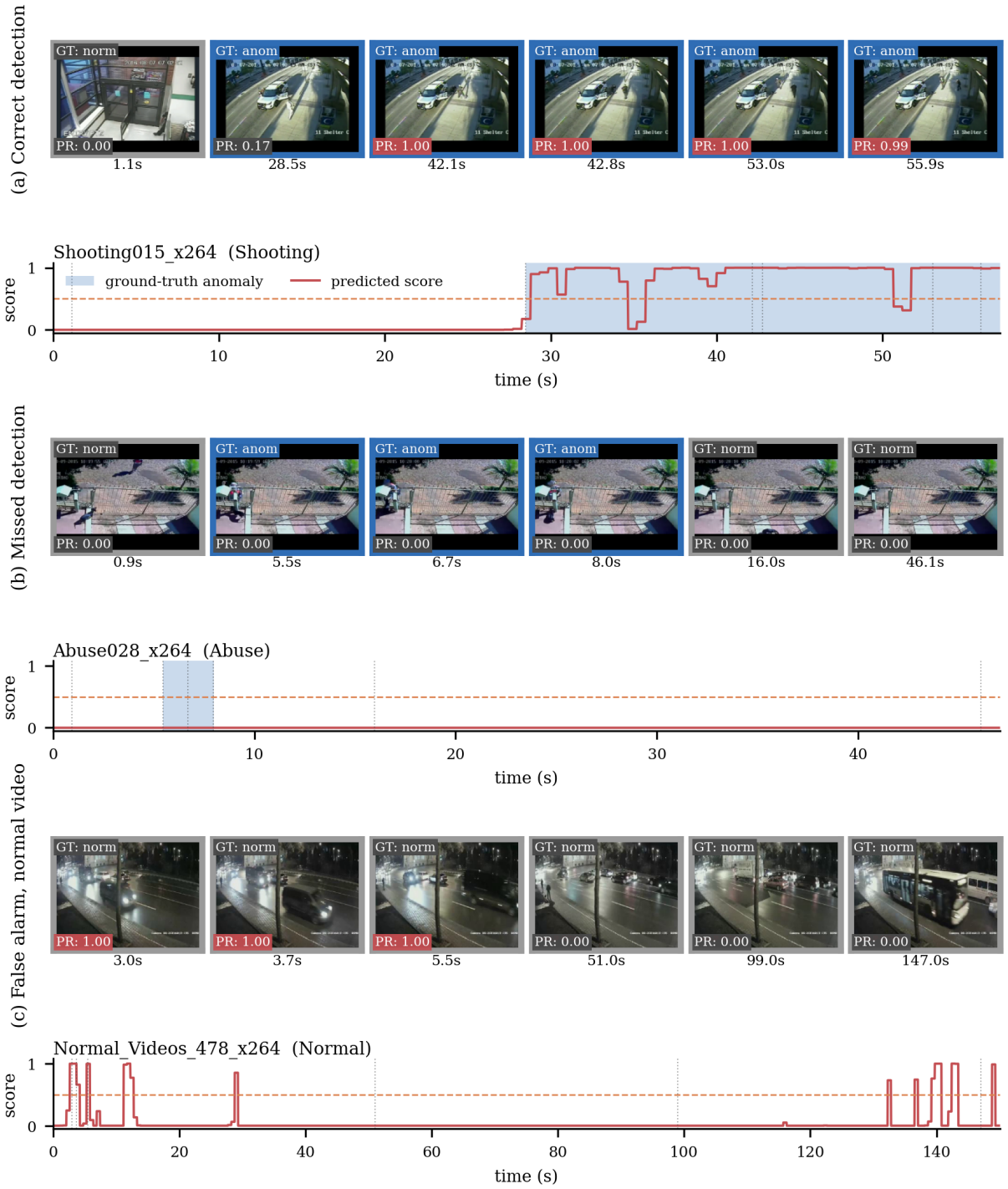


Figure 7: Qualitative results. Each block shows sampled frames above the score curve for one test video. Frame border and the upper tag give the ground-truth label (blue: annotated anomaly); the lower tag gives the predicted score (red when above threshold). (a) Shooting015: the score rises at the interval boundary and the annotated frames are scored 1.00. (b) Abuse028: frames inside the annotated interval at 5.5–8.0 s are scored 0.00; the event is missed entirely. (c) Normal_Videos_478: frames at 3.0–5.5 s are scored 1.00 although the video contains no annotated anomaly.

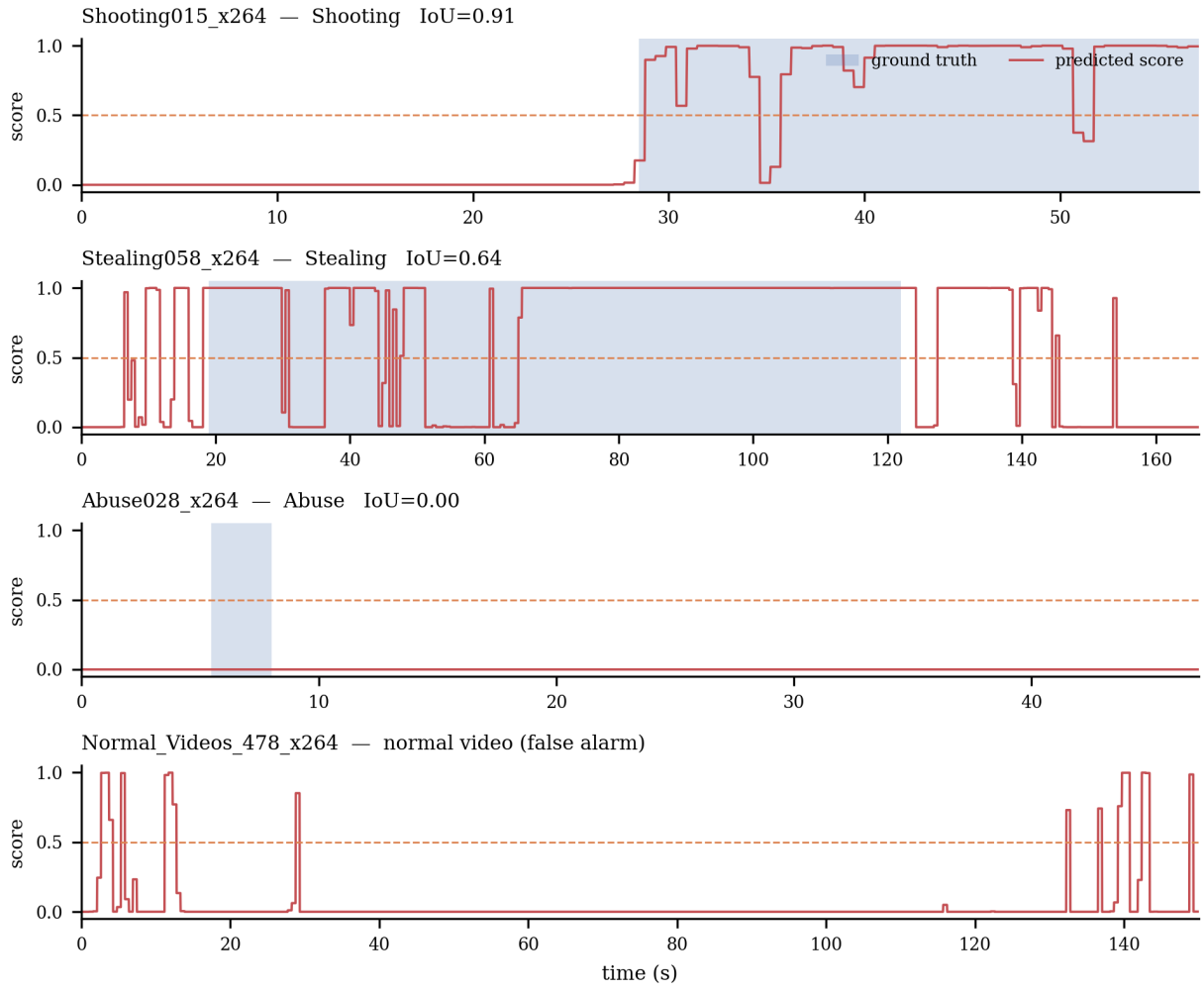


Figure 8: Predicted score against annotated intervals (shaded) for four test videos, ordered by temporal IoU.



Figure 9: One example per anomaly category. For each category we show the annotated frame that the detector scores highest, that is the best-case detection for that category, together with its score. Red border and score tag indicate a score above threshold. Twelve of thirteen categories contain at least one event detected at high confidence; Abuse does not. These panels show what the detector can localise, not how often it does so: aggregate per-category performance is given in Figure 10 and the distribution of temporal IoU in Section 6.

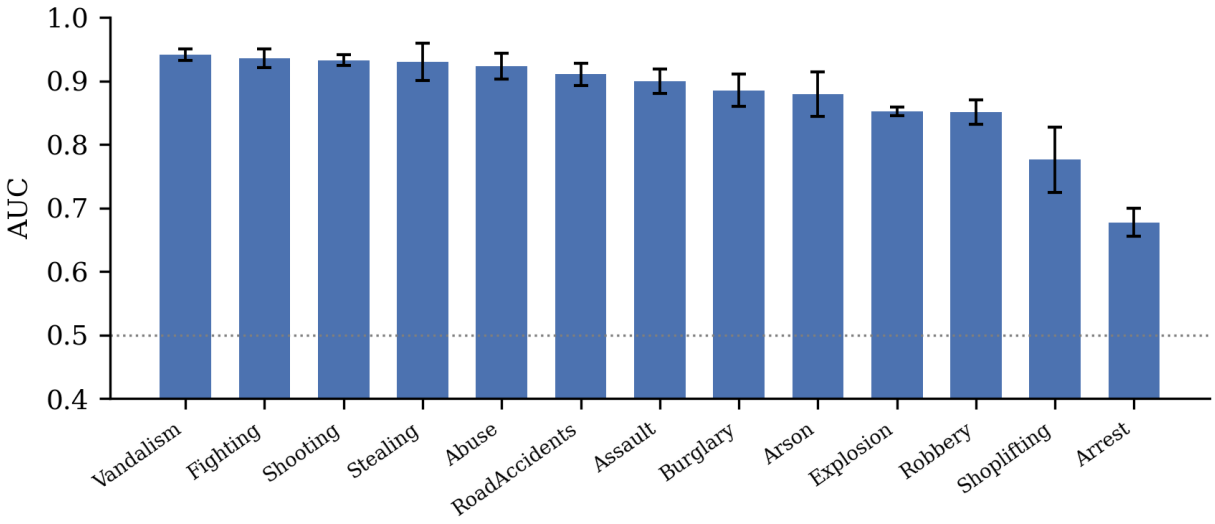


Figure 10: Per-category AUC, each category pooled with all normal test videos.

7 Discussion

7.1 What frame-level AUC measures

The mean-pooling row of Table 2 has the widest consequences. That model emits one score per video and assigns it to every frame, so it cannot localise by construction, yet it reaches 0.775 frame-level AUC – indistinguishable from the multiple-instance baseline at 0.777 and from the proposed model at 0.761. Ranking whole videos and localising events within them are different capabilities, and the metric on which this benchmark is ordinarily compared largely measures the first.

The localisation analysis gives the same conclusion from the other direction. At 0.769 frame-level AUC the median temporal IoU against the annotated interval is 0.000, and only 5 of 140 anomalous videos exceed 0.5. A model can rank frames well in aggregate while localising almost no individual event. The two measures that do separate the methods are the AUC restricted to anomalous videos and the fixed-threshold false-alarm rate: mean pooling scores 0.537 and 0.304 against 0.582 and 0.017 for ATA-HS. Only the first of the two survives measurement at a matched detection rate (Section 6.3).

7.2 What the benchmark can resolve

For our reproduced baseline under the standard protocol, a video-level bootstrap gives a 95% interval of $[0.696, 0.849]$. This is sampling uncertainty for one estimate on this test set rather than a fixed property of the benchmark, but its scale matters: it is an order of magnitude larger than the one- to two-point differences commonly reported between methods, so such differences are not resolvable here. This applies symmetrically. No pair of methods in Table 2 is separated on AUC: every paired bootstrap gives $p > 0.16$, and against mean pooling, which cannot localise at all, $p = 0.96$. It is why we do not claim an AUC improvement,. The published ordering reviewed in Section 2 additionally confounds detector with representation, since substituting CLIP features raises reported accuracy without changing the detection head; we therefore draw no conclusion from that ordering.

Seed variance compounds the problem. The proposed model varies by 0.024 across seeds against 0.005 for the baseline. A 2.66 M-parameter temporal model supervised by 1610 video-level labels is weakly constrained, whereas the baseline scores each snippet independently through a 1.07 M-parameter network and has less capacity to fit seed-specific structure. Because the ranking of ablations reverses between one and three runs, single-run ablation on this benchmark is not reliable at the magnitude of the effects being claimed.

Model selection is affected from a third direction. The validation signal available under weak supervision asks whether a video contains an anomaly; the test metric asks which frames are anomalous. Performance on one does not transfer to the other, and over 18 runs the two correlate at $r = -0.37$.

7.3 Where the proposed model does differ

At accuracy parity, ATA-HS lowers the false-alarm rate at threshold 0.5 from 0.023 for the baseline and 0.028 for dense self-attention to 0.017, with the smallest seed-to-seed variation of any row in Table 2. The effect survives removing the multi-scale branch, the hybrid heads, the adaptive budget and entmax individually, which attributes it to the sparse temporal formulation rather than to any single component. Over the 150 normal test videos, comprising 648,905 frames, the baseline raises 14,854 frames above that threshold and the full model raises 11,232.

What this is, and is not, is the substance of Section 6.3. The comparison holds the threshold fixed at 0.5 and so assumes the two models are comparably calibrated. They are not. Repeating it at a matched detection rate, an operating point invariant to the score scale, reverses the ordering: the baseline raises fewer false alarms at 95% detection (0.492 against 0.612) and

detects more anomalous frames at a 1% false-alarm rate (0.296 against 0.205). Mean pooling, worst on $\mathrm{FAR}_{0.5}$ by an order of magnitude, is best on the threshold-free measure.

The sparsity budget therefore buys calibration, not discrimination. It compresses most snippet scores toward zero, which places a large share of normal frames below any fixed threshold and makes 0.5 a usable default without tuning — a real convenience when deploying a detector, and the reason the released application needs no per-camera threshold calibration. It does not improve the separation of anomalous from normal frames, and we do not claim that it does. The broader point is that $\mathrm{FAR}_{0.5}$ has been read as a detection metric throughout this literature while measuring detection and calibration jointly, and our own initial reading of it was the same.

Table 4 isolates the selection mechanism. Entmax with a predicted budget is the most accurate and the least seed-sensitive of the four, ahead of dense attention, of entmax at a fixed temperature, and of non-differentiable top-K. Its realised support is 50.6% of snippets against 100% for softmax, so the sparsity is a measured property rather than one assumed from the transformation.

The motivating account of that gate is nonetheless wrong, and Section 6.4 reports the measurement that shows it. We proposed a per-video budget because event duration varies by a factor of thirty across categories; the predicted temperature in fact varies by four percent across the whole test set and is uncorrelated with annotated event duration. What the gate supplies is a schedule, dense early and sparse later, which a constant fixed at the converged value does not reproduce: that constant scores $0.682_{\pm 0.092}$ against $0.761_{\pm 0.014}$. We therefore claim automatic scheduling of sparsity rather than adaptation to content, and note that a mechanism which improves results for reasons other than those intended is worth reporting as such.

7.4 Secondary observations

The masked-normalisation result is a methodological caution. Centring the deviation and temporal heads over padded positions cost 0.096 AUC while *lowering* the training loss. The error is a train–test inconsistency rather than an optimisation failure: it is invisible in training diagnostics and would be attributed to the architecture unless tested for directly.

Two properties of the split explain much of the category head's uneven per-class recall. The training and test category balances differ (Figure 1a), so the categories with the most training data are the least tested; and several labels encode intent rather than appearance, which is not separable from surveillance imagery alone. Robbery, Stealing and Burglary differ in what the actor intends, not in what the camera records. At 0.379 top-1 accuracy over 13 classes the output is a ranking rather than a label, and the released application presents a ranked shortlist with probabilities.

The magnitude-selection row of Table 2 is the weakest and least stable. We attribute this to our simplification rather than to the published method, and note it as a limitation of the comparison rather than a finding about RTFM.

7.5 Recommendations

Four practices follow for work on this benchmark. Report seed variance: in our experiments the ranking of ablations reverses between one and three runs, and again between three and five. Test the mechanism a component is claimed to implement, not only whether the component helps; ours helped for a different reason than proposed, which only direct measurement revealed. Do not select models on video-level validation AUC, which does not predict the test metric. Report frame-level AUC alongside a localisation measure, since AUC near 0.77 is attainable with no localisation at all, and report false alarms at a matched detection rate rather than at a fixed threshold: the fixed-threshold rate moves with the scale of a model's scores, and in our table it ranks the methods in almost the reverse of the threshold-free order.

Limitations. Experiments use a single dataset, one backbone, and five seeds for the two models compared most closely and three for the remainder. The interval we report is specific to UCF-Crime's 290-video test set; larger test sets would narrow it. We did not evaluate vision-language features, which account for most recent reported gains. Confirming that the interval is a property of test-set size rather than of UCF-Crime specifically would require repeating the analysis on a larger benchmark such as XD-Violence [41], which contains 4754 videos.

8 Conclusion

We presented ATA-HS, a weakly-supervised video anomaly detector using differentiable sparse temporal attention with a per-video sparsity budget, together with an analysis of what the UCF-Crime benchmark can measure. ATA-HS reaches frame-level AUC a paired bootstrap cannot separate from a multiple-instance baseline, and lowers the false-alarm rate at the conventional fixed threshold from 0.023 to 0.017. We show that this last advantage is a property of the model's score scale rather than of its detection: at a matched detection rate the ordering reverses, and the method with the worst fixed-threshold rate has the best threshold-free one. The measurement analysis establishes that the benchmark's video-level confidence interval is ± 7.6 AUC points, that the validation signal available under weak supervision does not predict the test metric, and that frame-level AUC does not reflect temporal localisation. Code, trained model and application are released.

A Anomaly category classification

The detector answers whether an interval is anomalous, not of what kind. We additionally train a category head on the 810 anomalous training videos, which carry a video-level category label. Snippet features are pooled by attention rather than averaged, since the annotated event occupies a median 14% of its video, and the pooled descriptor is classified over the 13 categories.

Over the 140 anomalous test videos the head reaches 0.379 top-1 and 0.629 top-3 accuracy against a chance rate of 0.077. Per-class recall is uneven: RoadAccidents 0.74 and Arson 0.56, against Shooting 0.13 and Abuse, Arrest and Assault at 0.00.

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